Application of Multivariate Statistical Analysis in Risk Prediction and Efficacy Evaluation of Plasma Exchange in Hypertriglyceridemia-Induced Acute Pancreatitis

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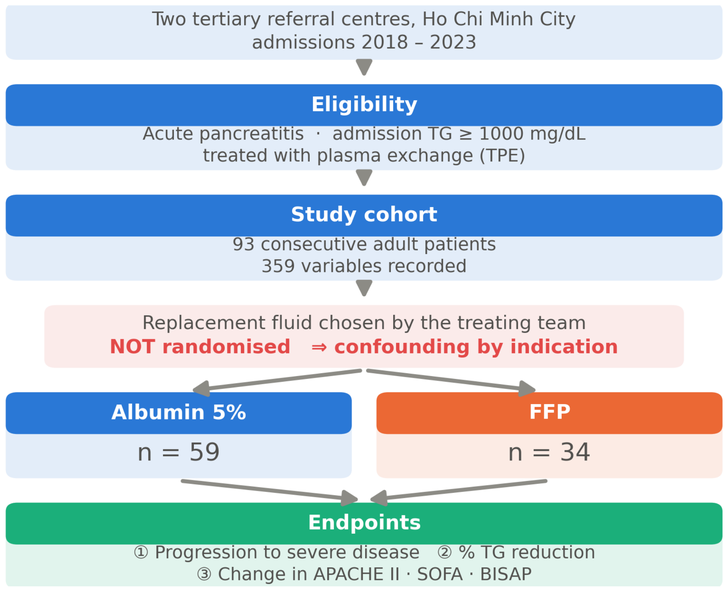
Clinical background and objective

Why HTG-induced pancreatitis matters. Hypertriglyceridemia (HTG) is the third most common cause of acute pancreatitis after gallstones and alcohol, and proportionally more frequent in Asian cohorts. Once serum triglyceride (TG) passes **1000 mg/dL** (11.3 mmol/L), chylomicrons obstruct the pancreatic microcirculation and lipase-released free fatty acids injure acinar cells. HTG-AP runs a more severe course than other aetiologies, so the aim is to bring TG below 500–1000 mg/dL *as fast as possible*.

Why the replacement fluid is an open question. Plasma exchange (TPE) removes chylomicrons directly, but ASFA grades HTG-AP only as **Category III, Grade 2C** [1]: the evidence establishes neither which patients benefit nor which fluid to use. **Fresh frozen plasma** restores lipoprotein lipase but carries transfusion reactions, citrate load and cost; **5% albumin** avoids all three. The choice is made at the bedside, *not* by protocol.

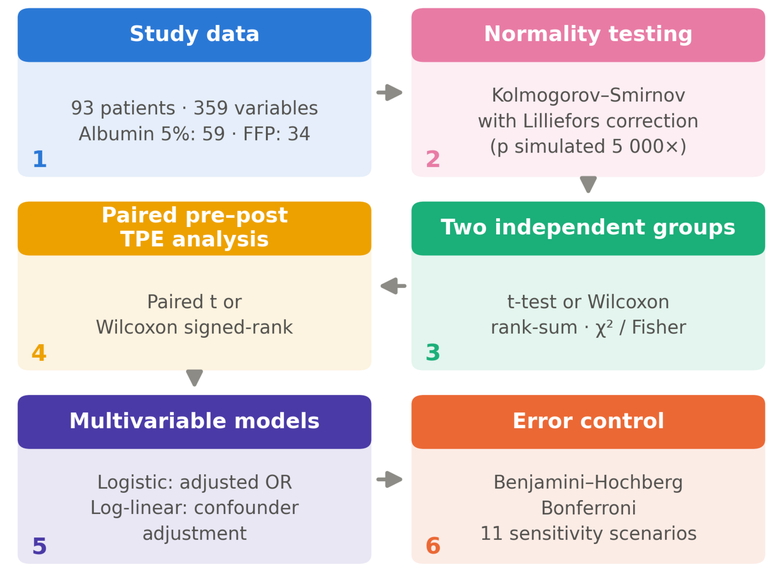
Objective. *Who deteriorates, does TPE work, does the fluid matter* — mapping onto risk prediction, paired pre–post measurement and group comparison under non-random allocation. With only **15 events** and **33 parallel tests**, **type I error control** and **sensitivity checking** separate a real effect from an apparent one.

Study design



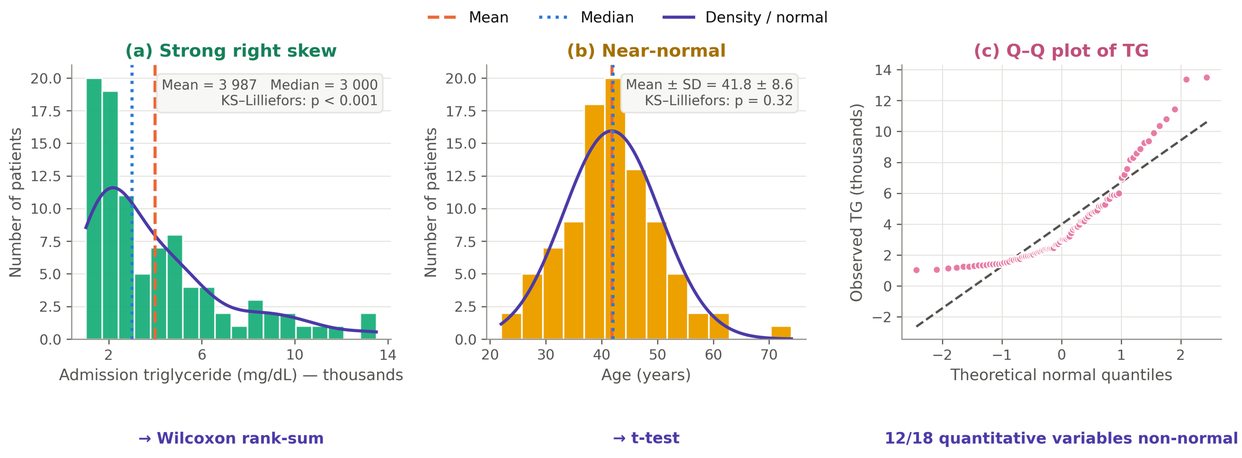
Patient flow. The branch point is the whole statistical difficulty: allocation was clinical, not random.

Statistical analysis pipeline



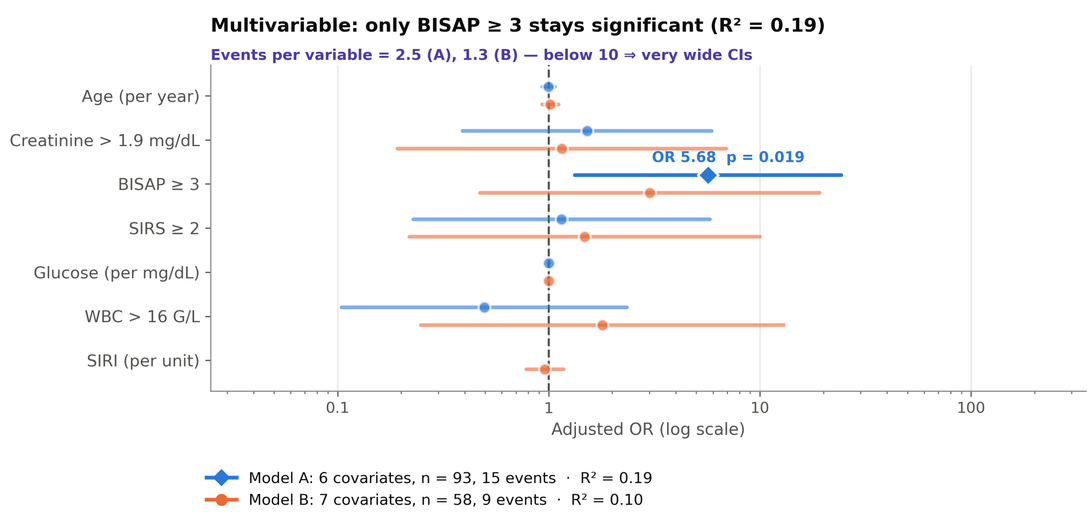
Tests follow from the normality result in step 2; steps 5 and 6 exist to keep false positives out of the conclusion.

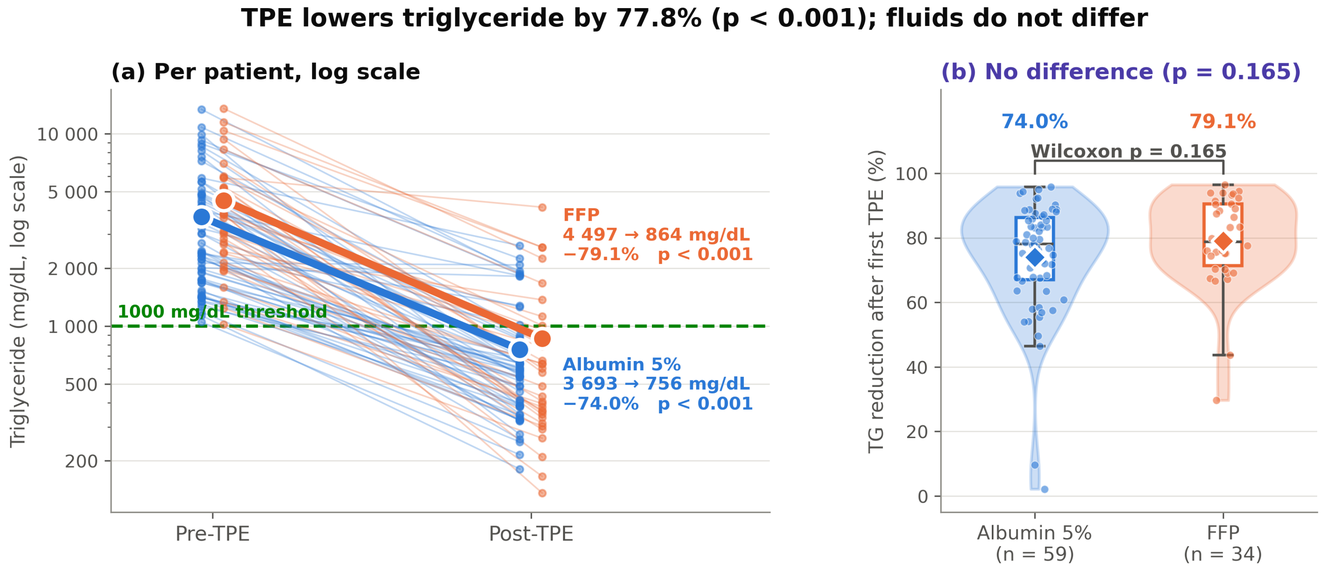
Result 1 — Distribution drives test choice · Univariate analysis of severity



► **12 of 18** quantitative variables are non-normal, so non-parametric tests are the default. Three factors have a 95% CI excluding 1, all with very wide intervals — a direct consequence of 15 events. **Caveat:** the severity proxy shares its components with BISAP and SIRS, so part of this association is **definitional**.

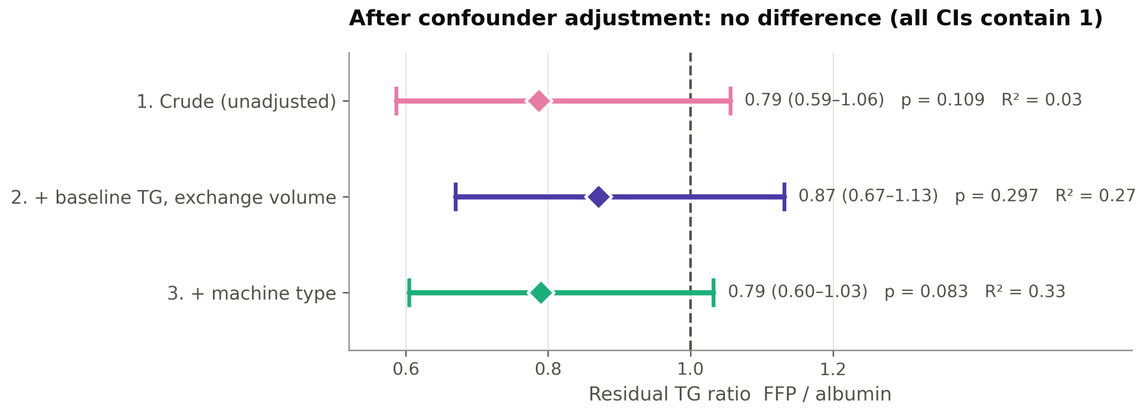
Result 2 — Multivariable logistic regression · Efficacy of plasma exchange (paired)

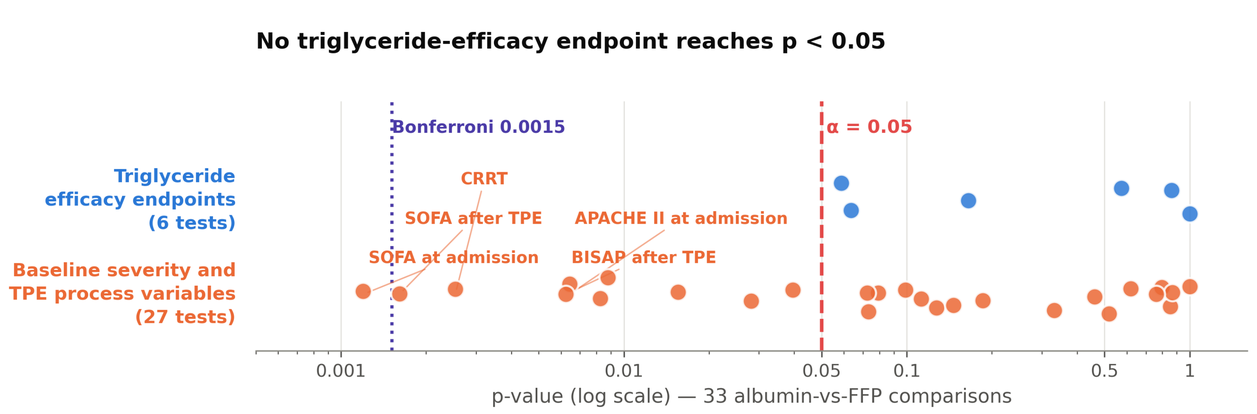




► Only BISAP ≥ 3 survives adjustment ($R^2 = 0.19$); at **2.5 events per covariate**, far below the Peduzzi rule of 10, the model is **exploratory**, not a prognostic score. The paired analysis, by contrast, is unambiguous: TG −77.8% ($p < 0.001$), **79.6%** reach < 1000 mg/dL after one session, and APACHE II, SOFA and BISAP all improve ($p < 0.001$).

Result 3 — Albumin 5% versus FFP: confounder adjustment · multiplicity control





► The FFP group was *sicker from the outset* (SOFA 5.6 vs 3.4 · APACHE II 20.1 vs 15.1 · CRRT 79% vs 48%; all $p < 0.01$) — **confounding by indication**. After adjustment the residual-TG ratio is 0.87 (0.67–1.13) and the reduction difference +4.1 points (−0.9 to 9.2): the correct conclusion is **"no difference detected"**, not "no difference".

Conclusion

In this non-randomised cohort of 93 patients, therapeutic plasma exchange produced a large and unambiguous fall in triglyceride (−77.8%, $p < 0.001$) together with improvement in APACHE II, SOFA and BISAP, consistently in both fluid groups — the one claim these data support without reservation. Risk prediction is far weaker: only BISAP ≥ 3 survived multivariable adjustment, on **2.5 events per covariate** and with a severity proxy that shares its components with its own predictors, so the model is **exploratory** rather than a prognostic score. The apparent advantage of FFP dissolved under scrutiny — the FFP group was sicker at baseline, and regression adjustment, multiplicity control [2] and eleven sensitivity scenarios all converged on **"no difference detected"** rather than "no difference". The contribution is therefore as much methodological as clinical: a **reproducible workflow** for small critical-care cohorts in which treatment is allocated by indication, where type I error control and sensitivity analysis are precisely what separate a real effect from an artefact. Propensity-score matching, mixed-effects models for the repeated measurements, and ultimately a randomised comparison of the two fluids are the natural next steps.

[1] Padmanabhan A. et al. *ASFA guidelines on therapeutic apheresis.* J Clin Apher 2019;34:171–354. [2] Benjamini Y., Hochberg Y. *Controlling the false discovery rate.* J R Stat Soc B 1995;57:289–300.

Sensitivity analysis

